applied sciences

MDPI

Type of the Paper (Article)

Accuracy Versus Computational Efficiency in Deep Learning for Land Use and Land Cover Classification: A Comparative Study of a Custom CNN and a Transfer-Learned ResNet50 Incorporating Grad-CAM Explainability and Deployment in Lahore, Pakistan.

Shahid Usman [1], Asif Nouman [1], Jamal Uddin [1], Waqar Mehmood [1], Ghani Rahman [2], and Khurrum Farid [3,*]

[1] Riphah International University; shahidusmaninfo@gmail.com, asif.nauman@riphah.edu.pk, waqar.mehmood@riphah.edu.pk
[2] University of Gujrat; ghani.rahman@uog.edu.pk
[3] Sheher Saaz Pvt Ltd; k.farid@shehersaaz.com,
* Correspondence: shahidusmaninfo@gmail.com; Tel.: (+923464500427)

Abstract

Accurate, high-resolution Land Use and Land Cover (LULC) mapping is crucial for effective urban planning and ecological monitoring in rapidly urbanizing areas. However, the trade-off between using lightweight models and deep learning models is not well quantified in terms of accuracy versus efficiency. This study compares a lightweight Custom 3-block CNN with a transfer-learned ResNet50 on the EuroSAT RGB dataset from Sentinel-2 imagery at 10-meter resolution. It evaluates classification accuracy, per-class discriminative ability, computational efficiency, and field deployment in Lahore, Pakistan semi-arid, fast-growing city facing serious environmental issues. On the EuroSAT benchmark, Custom CNN achieves 81.38% overall accuracy and an 80.50% Macro F1-score, whereas ResNet50 reaches 97.41% accuracy and a 97.32% Macro F1-score, offering a 16.03 percentage-point accuracy increase at 25.7 times the computational cost. Grad-CAM analysis confirms that ResNet50's classifications are based on meaningful geographic features, such as edge continuity for highways, recurring structures for residential zones, and spectral uniformity for forests. It also identifies interpretability issues caused by spectral confusion between sediment and turbidity, and phenological differences. Deploying ResNet50 on 4,289 Sentinel-2 patches over Lahore shows a land use distribution of 23.25% Urban, 45.57% Agricultural, 29.32% Natural vegetation, and 1.85% Water, with Forest at 0.00%, indicating near-total loss of dense tree cover and supporting documented urban expansion at 10-meter resolution. These findings offer a reproducible, automated benchmark for city land monitoring and introduce a scalable, explainable deep learning framework adaptable to other semi-arid, rapidly growing cities with similar environmental challenges.

Keywords: land use and land cover (LULC); convolutional neural networks; ResNet50; transfer learning; EuroSAT; Sentinel-2; Grad-CAM; explainable artificial intelligence; urban sprawl; computational efficiency.

Academic Editor: Firstname Lastname

Received: date
Revised: date
Accepted: date
Published: date

Citation: To be added by editorial staff during production.

Copyright: © 2025 by the authors. Submitted for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

40